Find the net force before dividing by mass

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Net force / ES: fuerza neta / PT: força resultante

Combine opposite forces: subtract and keep the direction of the larger one.

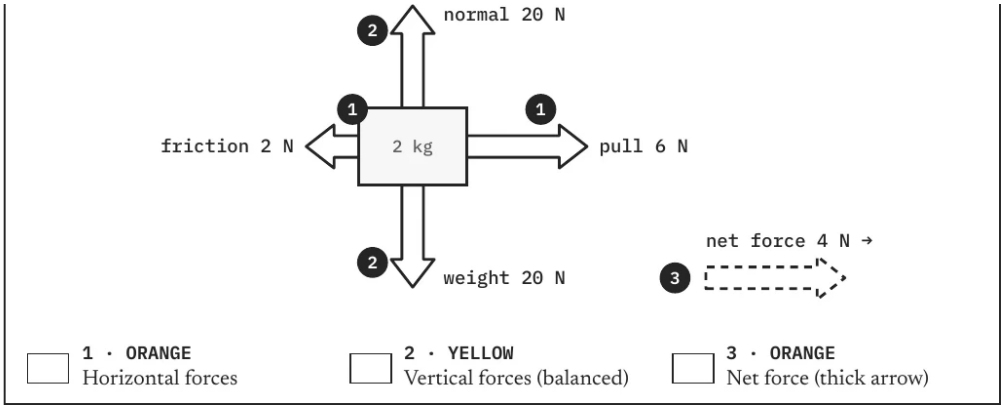
Also on your note page: Newton's second law, Direction, Balanced.

B READ THE RELATIONSHIP

Net force is the sum of all the forces on one object, with directions included.

Add forces first. Then divide by mass.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 4 kg cart has 12 N pulling right and 4 N of friction left. Find the acceleration.

Given m = 4 kg 12 N right 4 N left

Rule a = F net ÷ m

Work F net = 12 − 4 = 8 N right a = 8 ÷ 4

Answer 2 m/s² right with the net force

E SAVE FOR THE FINAL MEETING

F = ma links it all: from XJ's mass and the net force on it, we can calculate its acceleration, and from that, how its velocity will change.

Read the physics: Net force

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Choose one object and a positive direction. Write forces to the right and left with opposite signs. Add before you divide by mass.

02 HOW IT WORKS

A net-force arrow summarizes the individual forces. Do not add that arrow again. Divide the signed total by mass. Acceleration points in the direction of the net force.

03 WHAT TO CHECK

Velocity can point the other way. A left-moving cart with a rightward acceleration slows at first.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 4 kg cart has 12 N pulling right and 4 N of friction left. Find the acceleration. 2 m/s² right with the net force.

05 CONNECT TO XJ

F = ma links it all: from XJ's mass and the net force on it, we can calculate its acceleration, and from that, how its velocity will change.

SAY IT BEFORE YOU WRITE

Which two steps give acceleration? I added ___ and ___ to get ___. Then I divided by ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words	How can you tell?
How would you explain net force to a friend? In my words, this means ___.	Two people push one cart in opposite directions. Why do the arrow directions matter? I can tell because ___.
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both? My example shows ___.	A student adds those pushes to get 10 N right. Which push did they count the wrong way? I would change ___ because ___.

Newton's second law

ES: segunda ley de Newton / PT: segunda lei de Newton

Meaning in my words	How can you tell?
What does Newton's second law tell us about a cart? In my words, this means ___.	Why add the forces before dividing by mass? I can tell because ___.
Example or sketch	**Non-example and why**
A 2 kg cart feels 7 N right and 3 N left. What acceleration do you predict? My example shows ___.	A partner uses 7 N alone to find acceleration. Which part of the diagram contradicts that? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Direction

ES: dirección y sentido / PT: direção e sentido

Meaning in my words	How can you tell?
How would you explain direction to a friend? In my words, this means ___.	A cart moves right while its net force points left. Which way does acceleration point? I can tell because ___.
Example or sketch	**Non-example and why**
Draw arrows for a cart moving left and slowing down. My example shows ___.	"The biggest force points east, so the cart must be moving east." What can we actually tell if the net force is east? I would change ___ because ___.

Balanced

ES: fuerzas equilibradas / PT: forças equilibradas

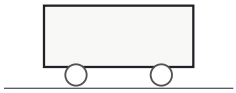
Meaning in my words	How can you tell?
What does it mean to say the forces on a cart are balanced? In my words, this means ___.	A cart feels 5 N left and 5 N right. How can you tell its horizontal forces balance? I can tell because ___.
Example or sketch	**Non-example and why**
A book rests on a table. What two vertical arrows would you draw? My example shows ___.	One cart pushes another, and the second pushes back. Do these two forces balance on the first cart? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw an object with two opposing forces. Write the signed sum under the net arrow.

Labels you can use: force · sum · acceleration.



Tell a partner

I added ___ and ___ to get ___. Then I divided by ___.

Moving left. Pushed right.

13

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish motion, net force, and acceleration.

CHOOSE ONE JOB TOGETHER

CONTROLS ENGINEER ☐

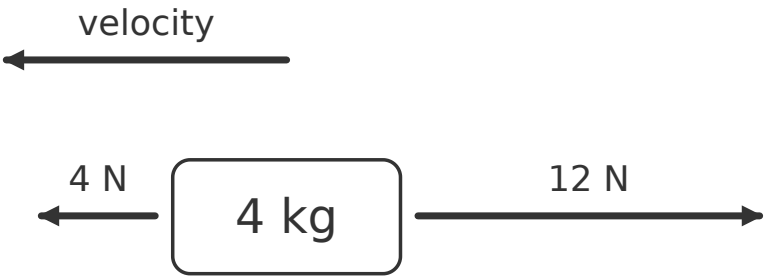
Draw the net-force and acceleration arrows. Tell the operator what happens to speed while the body still moves left.

REPLAY ANALYST ☐

Label separate arrows for motion and acceleration. Explain to viewers how they can point opposite ways.

Word help: net force / acceleration / slowing down

A tracking model moves left. Its **4 kg** body feels **12 N** right and **4 N** left. The status display has separate motion and force arrows.



01 OUR FIRST CALL

What happens to speed at first? Include the net force and acceleration in your call.

It moves ___ while acceleration points ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the updated display. What does a force arrow tell you, and what does it not tell you?

Now we would say ___ because ___.

Switch speakers. Can motion and acceleration point in opposite directions?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks